

# SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

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## COURSE OUTCOME COVERED

CO5: Design Pushdown Automata and analyze their equivalence with Context-Free Grammars through formal construction and proof techniques. ( BL :3)

Assessment Tool Weightage: 40%

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QUESTIONS: 5  
Duration : 1 Hour

Total Marks:25

## Constraint Based Problem

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### Code of Conduct:

I A Manoj Reddy 192311171 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



## SIMATS ENGINEERING ASSESSMENT TOOLS

1. Design a Pushdown Automaton that accepts the language

$$L = \{ a^n b^n \mid n \geq 1 \}$$

**Constraint:** You are allowed to use only one stack symbol (besides initial symbol)

2. CFG to PDA Conversion with Limited Productions

Given a Context-Free Grammar:

$$S \rightarrow aSb \mid ab$$

Construct an equivalent PDA.

**Constraint:** PDA must be constructed using only 2 states

3. PDA to CFG Conversion with Restricted Variables

Convert a given PDA into an equivalent CFG.

**Constraint:** The resulting CFG must use at most 3 variables (non-terminals)

4. Turing Machine with Limited Tape Movement

Design a Turing Machine to recognize the language

$$L = \{ ww \mid w \in \{0,1\}^* \}$$

**Constraint:** The tape head can move only to the right (no left movement allowed)

5. Consider a language  $L = \{ a^n b^n c^n \mid n \geq 1 \}$

**Constraint:** Design a Turing Machine for L

PDA for  $L = \{ a^n b^n \mid n \geq 1 \}$

Constraint: only one stack symbol besides the initial stack symbol

$$\Gamma = \{ z_0, x \}$$

Use three states:

$$Q = \{ q_0, q_1, q_f \}$$

transitions

$$\delta(q_0, a, z_0) = (q_0, x z_0)$$

$$\delta(q_0, a, x) = (q_0, x x)$$

when the first  $b$  appears, move to  $q_1$  and

Pop one  $x$ !

$$\delta(q_0, b, x) = (q_1, \epsilon)$$

for every remaining  $b$ , pop one  $x$ :

$$\delta(q_1, b, x) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_f, z_0)$$

for input:

$aabb$

the operation is

$$(q_0, aabb, z_0)$$

$$\vdash (q_0, abb, x z_0)$$

$$\vdash (q_0, abb, x x z_0)$$

$$\vdash (q_0, b, x x z_0)$$

$$\vdash (q_1, \epsilon, z_0)$$

$$\vdash (q_f, \epsilon, z_0)$$

2.

Convert CFG  $S \rightarrow aSb \mid ab$  into a 2-state PDA

$$P = (\{q_0, q_f\}, \{a, b\}, \{S, a, b, \epsilon\}, \delta, q_0, \epsilon, q_f)$$

Transitions:

$$\delta(q_0, \epsilon, \epsilon) = (q_0, S\epsilon)$$

$$S \rightarrow aSb$$

$$\delta(q_0, \epsilon, S) = (q_0, aSb)$$

$$S \rightarrow ab$$

$$\delta(q_0, \epsilon, S) = (q_0, ab)$$

$$\delta(q_0, a, a) = (q_0, \epsilon)$$

$$\delta(q_0, b, b) = (q_0, \epsilon)$$

thus the PDA has exactly:  $\delta(q_0, \epsilon, \epsilon) = (q_f, \epsilon)$

2 states:  $q_0, q_f$

and accepts the same language

$$L = \{a^n b^n \mid n \geq 1\}$$

3.

PDA to CFG with at most 3 variables

$$S \rightarrow aSb \mid ab$$

$$V = \{S\}$$

$$1 \leq 3$$

for example:

$$S \rightarrow aSb$$



$$S \Rightarrow aSb$$

$$\Rightarrow aaaSbb$$

$$\Rightarrow aaaS(ab)bb$$

$$\Rightarrow aaaSbbb$$

$$L(a) = \{ a^n b^n \mid n \geq 1 \}$$

4) TM for  $L = \{ ww \mid w \in \{0,1\}^* \}$  with only right movement:-

This is impossible with a standard single-tape deterministic TM

$$L = \{ ww \mid w \in \{0,1\}^* \}$$

example: of  $L$ :

$$00 = 0 \cdot 0$$

$$11 = 1 \cdot 1$$

$$0101 = 01 \cdot 01$$

$$1010 = 10 \cdot 10$$

$$0010 \notin L$$

A normal Turing machine could make symbols in the first half and repeatedly move between the two halves to compare corresponding symbols under the given constraint.

Head movement = R only

No standard one-tape TM satisfying the stated constraint exists.

5. TM for  $L = \{ a^n b^n c^n \mid n \geq 1 \}$

Use markers:-

$a \rightarrow x$

$b \rightarrow y$

$c \rightarrow z$

### Algorithm

- 1) Start the left end
- 2) find the first unmarked 'a' and replace it with x
- 3) Move right until the first unmarked 'b' is found  
replace it with y.
- 4) Continue right until the first unmarked 'c' is  
found ; replace it with z.
- 5) move left to the ~~beginning~~
- 6) Repeat the process.
- 7) where there are no unmarked 'a's' check,  
whether any unmarked 'b' or 'c' remain.
- 8) if only x, y/z remain, ACCEPT.
- a) Otherwise, REJECT.

example: aabbcc

accept

$L = \{ a^n b^n c^n \mid n \geq 1 \}$

# SIMATS ENGINEERING ASSESSMENT TOOLS

## RUBRICS FOR CALCULATION

| Criteria                 | Weightage | Level 4 – Exemplary (5 Marks)                                | Level 3 – Proficient (4 Marks)         | Level 2 – Developing (2–3 Marks)             | Level 1 – Beginning (0–1 Mark)  |
|--------------------------|-----------|--------------------------------------------------------------|----------------------------------------|----------------------------------------------|---------------------------------|
| Conceptual Understanding | 30        | Demonstrates clear and complete understanding of the concept | Good understanding with minor gaps     | Basic understanding with some misconceptions | Poor or incorrect understanding |
| Accuracy of Answer       | 25        | Completely correct answer with no errors                     | Mostly correct with minor errors       | Partially correct with noticeable errors     | Incorrect or irrelevant answer  |
| Application of Knowledge | 20        | Correctly applies concepts to solve the question/problem     | Applies concepts with minor mistakes   | Limited or partially correct application     | No or incorrect application     |
| Completeness of Response | 15        | Covers all required parts of the question                    | Covers most parts with minor omissions | Covers only some parts                       | Incomplete or missing response  |
| Clarity & Presentation   | 10        | Clear, concise, and well-organized answer                    | Understandable with minor issues       | Somewhat unclear or poorly structured        | Difficult to understand         |

86/100

*[Signature]*